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Germline Small-Variant Annotation and Pathogenicity Classification on NCHC Draft 2026

A VEP-based annotation and point-based in-house ACMG classification workflow for germline small variants

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Reference build GRCh38 • **Ensembl VEP** v115 • **MANE** v1.4 • **ACMG point-base** (SVC4 draft) | 2026

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A VEP-based annotation and point-based in-house ACMG classification workflow for germline small variants — GRCh38.

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1. Summary

This workflow provides standardized annotation and automated pathogenicity classification for germline small variants on the National Center for High-performance Computing (NCHC) environment. It accepts a **GRCh38** germline small-variant VCF produced by either GATK or DRAGEN, performs Ensembl VEP annotation on MANE transcripts, and applies an automated ACMG/AMP classification using a **point-based** scheme adapted from the ACMG/AMP/CAP/ClinGen Sequence Variant Interpretation (SVC) v4 draft. It succeeds the previous, rule-based GenDiseak (GDK) platform, and is designed to be team-accessible, reproducible, and maintainable.

Benchmarked against ClinVar (review status ≥ 2 stars, N = 629,412), the workflow reproduces expert classifications with high concordance — **99.6%** for benign/likely-benign and **97.5%** for pathogenic/likely-pathogenic variants — and, unlike the rule-based predecessor, resolves roughly a quarter of ClinVar variants of uncertain significance (VUS) toward a definitive class where the evidence supports it.

2. Background & Motivation

2.1 Annotation drives classification

Variant interpretation under the ACMG/AMP framework depends heavily on annotation. Different annotation tools can assign different HGVS descriptions and coding-impact consequences to the same variant; even on the same transcripts, based on our previous study, different annotation tools reach a concordance of only about 58.5% for HGVSc and 85.6% for coding impact, which in turn affects downstream ACMG classification. Standardized transcript sets (MANE) and a single, well-maintained annotation engine are therefore prerequisites for reliable, reproducible classification.

2.2 Why a new workflow

The legacy GDK platform (rule-based combining) will no longer be actively maintained. To keep analysis running and shareable across the team, the pipeline was rebuilt on NCHC and modernized:

- **GRCh38** + **Ensembl VEP** — actively maintained, a rich plugin ecosystem, and alignment with current databases.
- **MANE Select** + **MANE Plus Clinical** transcript annotation for standardized, clinically relevant reporting.
- **Point-based ACMG classification** adapted from the SVC v4 draft — reducing conflicting evidence combinations, enabling VUS sub-classification (e.g., VUS-High) for triage, and adding prediction scores absent from the previous ANNOVAR/GDK flow.

3. Annotation Pipeline

Pipeline overview: FASTQ (sequencing) → GATK / DRAGEN variant calling (GRCh38) → small-variant VCF (SNV / indel / MNV) → VEP annotation (MANE Select + Plus Clinical) → in-house ACMG point-based class → annotated TSV → filter · review.

3.1 Input and VCF preprocessing

The workflow accepts a germline small-variant VCF (or `.vcf.gz`) generated on GRCh38 by GATK or DRAGEN; the upstream calling process is unchanged. Preprocessing standardizes the input in four steps:

1. Retain only chromosomes 1–22, X, Y, and M (MT).
2. Normalize (split multiallelic sites, left-align) and remove duplicate sites.
3. Keep only SNV / indel / MNV; exclude non-standard ALT alleles and structural variants.
4. Split by chromosome and extract the 65 MANE Plus Clinical transcript regions (with a 5,000 bp buffer) for dedicated handling.

3.2 Transcript model — MANE

Annotation uses the Matched Annotation from NCBI and EMBL-EBI (MANE) transcript set — a genome-wide set of representative transcripts agreed between RefSeq and Ensembl/GENCODE. This workflow uses **MANE v1.4** on GRCh38, applying both MANE Select and MANE Plus Clinical; 65 genes carry both a MANE Select and a MANE Plus Clinical transcript. Transcript selection priority differs by gene set so that clinically relevant transcripts are preferred where they exist.

Table 1. Transcript selection order.

Variant set	Selection priority
All variants (general)	mane_select, mane_plus_clinical, rank, canonical, appris, biotype, length
Genes with both MANE Select + Plus Clinical (65)	mane_plus_clinical, mane_select, rank, canonical, appris, biotype, length

3.3 Annotation — Ensembl VEP v115

Annotation is performed with Ensembl VEP v115, combining a set of plugins with custom database annotations.

Table 2. VEP plugins and custom annotation sources.

Category	Sources
Splicing predictions	SpliceAI, dbscSNV, MaxEntScan
Pathogenicity predictions	dbNSFP, PrimateAI, LoFtool
Gene tolerance to variation	pLI, LOEUF, DosageSensitivity
Transcript / NMD	NMD (nonsense-mediated decay escape)
Phenotype & citations	satMutMPRA
Custom databases	RepeatMasker, DVD (v9.2), MitoMap, ClinVar, gnomAD, TWB, SG10K

3.4 Runtime and storage

As a whole-genome (WGS) reference point, sample HRD441 (original N = 4,864,719 variants; final N = 4,932,880 after normalization) completed in approximately 920 CPU-hours with a peak storage footprint of 4.7 GB (TSV ≈ 3.5 GB).

Table 3. Stage-level resource profile (WGS, HRD441).

Stage	Cores / memory	Wall-clock time
Preprocess	2 / 13 GB	4 m 05 s
Annotate (per chromosome)	14 / 92 GB	8 s – 5 h 04 m 09 s
Merge & generate TSV	1 / 7 GB	5 m 16 s
Total	≈ 920 CPU-hours	Peak storage 4.7 GB

4. Pathogenicity Classification

4.1 From combining rules to points

The classifier is built on the ACMG/AMP 2015 five-tier system (Pathogenic, Likely Pathogenic, VUS, Likely Benign, Benign) with its evidence codes. Rather than the 2015 discrete combining rules, it aggregates evidence on a point scale adapted from the SVC v4 draft (Tavtigian 2020): Very Strong **+8**, Strong **+4**, Moderate **+2**, Supporting **+1**, with benign criteria taking the corresponding negative points. The aggregated score maps to the five tiers as follows: Benign ≤ **−4**, Likely Benign **−3 to −1**, Uncertain Significance (VUS) **0 to 5**, Likely Pathogenic **6 to 9**, and Pathogenic ≥ **10**. Point aggregation reduces conflicting combinations and allows VUS sub-classification. The predecessor GenDiseak (GDK) was rule-based; the NCHC classifier — like Varsome — is point-based.

> **NOTE** The scheme is adapted from the SVC v4 draft; it is not a full implementation of the final v4 specification. Exact point thresholds for the final class are defined in the tool configuration (`utils/config.json`) and should be cited from there for any formal report. In `config.json` the tunable items are the **CADD, DANN, SpliceAI, LOEUF, and gnomAD allele-frequency (AF) thresholds** and the **file paths** of the reference resources; the point/score scheme itself is adjusted separately in **`utils/rescore.py`** .

4.2 Implemented evidence criteria

The classifier evaluates the following criteria automatically from annotation and reference data. Criteria that require clinical, family, functional, or case-level information are not automated and remain for manual review.

Table 4. Automated ACMG criteria, implemented strength, and evidence used.

Criterion	Evidence used	Logic (summary)
PVS1 (VeryStrong / Strong)	VEP consequence, LOEUF, NMD, LoF gene list	Null variant (stop-gain, frameshift, splice donor/acceptor) in a gene where LoF is a known mechanism; LOEUF ≤ 0.6; NMD-escape logic modulates strength.
PS1 / PM5	pathogenicDB (VEP-annotated ClinVar)	Same amino-acid change (PS1) or same residue, different change (PM5) as an established pathogenic variant, on the same transcript.
PS3	GoF list (ClinVar + HGMD)	Variant present in a curated gain-of-function set supportive of a damaging functional effect.
PM1	pathogenicDB region	≥ 4 pathogenic variants within ±25 bp; mitochondrial variants excluded per ClinGen.
PM2 / BA1 / BS1 / BS2	MOI + gnomAD frequency	Frequency thresholds conditioned on mode of inheritance; see Table 6. QC: AN > 2000, coverage > 20, FILTER = PASS.
PM4 / BP3	VEP consequence, RepeatMasker	Protein-length change (in-frame indel / stop-loss) in a non-repeat (PM4) vs repeat (BP3) region.
PP3 / BP4 / BP7	SpliceAI, CADD, DANN	Computational support for (PP3) or against (BP4) impact; BP7 for synonymous variants with no predicted splice effect. See Table 6.
PP5 / BP6 (scaled)	ClinVar, DVD, MitoMap	Reputable-source classification; strength scales with ClinVar review status (0–4 stars).

> **REQUIRES MANUAL REVIEW** PS2 / PM6 (de novo), PS4 (case–control), PM3 (in trans), PP1 / BS4 (segregation), PP2 / BP1 (gene spectrum), PP4 (phenotype), and BP2 / BP5 (allelic / alternate cause) depend on clinical, family, or functional data and are not assigned automatically.

4.3 Reference data for interpretation

Reference resources are built and versioned so that each assignment is traceable.

Table 5. Curated reference data underpinning the automated criteria.

Resource	Construction / source
LoF gene list (PVS1)	ClinVar 20251109 P/LP, ≥2★, LoF molecular consequence (frameshift, nonsense, splice donor/acceptor); genes with > 2 qualifying variants.
pathogenicDB — AA change (PS1/PM5)	VEP-annotated ClinVar P/LP, ≥2★, missense; keyed by transcript, amino-acid change and position.
pathogenicDB — region (PM1)	ClinVar P/LP, ≥2★, missense / in-frame indel / start-loss; mitochondrial variants excluded (ClinGen).
GoF list (PS3)	Aggregated from ClinVar and HGMD.
Mode of inheritance (PM2/BS2)	Aggregated from CGD, ClinGen, and PanelApp.
Population frequency	gnomAD (exome & genome); QC AN > 2000, coverage > 20, PASS.
BA1 exception list	ClinGen BA1 exception variants excluded from stand-alone benign.
Reputable source (PP5/BP6)	ClinVar (CLNSIG, CLNREVSTAT), DeafnessVD (DVD v9.2), MitoMap.

4.4 Selected decision thresholds

Table 6. Representative thresholds for frequency- and score-based criteria.

Criterion	Threshold
PM2	MOI = AD/XL/YL/blank & AF ≤ 1.44×10^{-5}; or MOI = AR & AF ≤ 1×10^{-4} or nhomalt ≤ 2 (absent/rare after QC).
BA1 / BS1	gnomAD AF ≥ 0.05 (BA1, unless in ClinGen exception list); AF ≥ 0.01 (BS1).
BS2	Observed in healthy adults at frequency above the MOI-conditioned PM2 threshold.
PP3	SpliceAI DS (AG/AL/DG/DL) ≥ 0.2; or CADD-phred ≥ 25.3 (Supporting) / ≥ 28.1 (Moderate); or DANN rankscore ≥ 0.999 when CADD is null.
BP4	SpliceAI DS_AG/AL/DG/DL ≤ 0.1 (all) **AND** CADD-phred ≤ 22.7 (Supporting) / ≤ 17.3 (Moderate) / ≤ 0.15 (Strong); or, when CADD is null, DANN rankscore ≤ 0.974 (Supporting) / ≤ 0.915 (Moderate) / ≤ 0.478 (Strong) with the same SpliceAI condition.

Criterion	Threshold
BP7	Synonymous variant **AND** SpliceAI DS_AG/AL/DG/DL ≤ 0.1 (all); no predicted splice effect.
PP5 / BP6	Strength scaled by ClinVar review status: 0–1★ Supporting → 2★ Moderate/Strong → 3–4★ Strong/Very Strong.

4.5 Rule interaction (avoiding double-counting)

Related criteria are disabled by design so that the same underlying evidence is not scored twice.

Table 7. Disable logic between criteria.

Trigger	Disables	Rationale
BA1 / PM2	BS1, BS2	Frequency evidence not counted twice across benign strong and moderate pathogenic.
PVS1	PP3, PM4, BP4	LoF / splice and length-change evidence already captured by PVS1.
PM1	BP3; (MT variants)	Region evidence supersedes BP3; PM1 disabled for mitochondrial variants (ClinGen).
PM4	PP3	Length-change and computational evidence overlap.
(MT variants)	BP3, PM1	Mitochondrial-specific exclusions per ClinGen guidance.

5. Benchmark

The classifier was benchmarked against ClinVar (release 20251109; composition at review status ≥ 2 stars, N = 629,412: B/LB 271,884; P/LP 84,611; VUS 272,472; other 108). Performance was evaluated on the updated configuration and separately for two ClinVar confidence levels (review status ≥ 2★ and ≥ 3★), with the reputable-source criteria **PP5/BP6 either included or excluded**, to show their effect on recall.

Table 8. NCHC classification performance vs ClinVar, by configuration. Footnote: based on the updated version — LoF gene list and LOEUF ≤ 0.6.

Panel A — Benign / Likely-benign (B/LB)

Metric	Exclude PP5/BP6 (≥3★)	Exclude PP5/BP6 (≥2★)	Include PP5/BP6 (≥3★)	Include PP5/BP6 (≥2★)
Precision	0.8505	0.7507	0.9040	0.8197
Recall	0.6470	0.6637	0.9996	0.9962
Accuracy	0.8698	0.7555	0.9715	0.9028
F1	0.7349	0.7045	0.9494	0.8993

Panel B — Pathogenic / Likely-pathogenic (P/LP)

Metric	Exclude PP5/BP6 (≥3★)	Exclude PP5/BP6 (≥2★)	Include PP5/BP6 (≥3★)	Include PP5/BP6 (≥2★)
Precision	0.9646	0.9669	0.9664	0.9657
Recall	0.9260	0.9109	0.9991	0.9738
Accuracy	0.9316	0.9787	0.9795	0.9910
F1	0.9449	0.9381	0.9825	0.9697

> **KEY RESULTS** Including the reputable-source criteria (PP5/BP6) sharply raises recall (B/LB recall 0.65 → ≈1.00; P/LP recall 0.93 → ≈1.00) at little precision cost, giving the best F1 — B/LB 0.949 and P/LP 0.983 at ≥3★. At ≥2★ (more variants, lower average review confidence) F1 is modestly lower (B/LB 0.899, P/LP 0.970). Excluding PP5/BP6 mainly costs B/LB recall, reflecting how many benign calls lean on reputable-source concordance.

The comparison below highlights the behavioural difference between the point-based NCHC classifier, the conservative rule-based GDK, and the more aggressive Varsome. For an equal-footing comparison, Varsome's VUS-toward-B/LB and VUS-toward-P/LP calls are collapsed back to VUS (VUS-BLB/PLP → VUS), and all three tools are scored on the matched ClinVar subset (N = 626,562).

Table 9. Behaviour comparison against the same ClinVar reference (N = 626,562).

Metric	NCHC	GDK	Varsome (VUS-BLB/PLP → VUS)
ClinVar B/LB reproduced	99.6%	86.8%	99.8%
ClinVar P/LP reproduced	97.5%	81.7%	98.0%
ClinVar VUS resolved to a definitive class	23.0%	0%	55.9%

6. Using the Workflow

The workflow is split into two repositories — annotation and classification:

- **Annotation:** https://github.com/leechiehyu/VEP-small_variant
- **Classification:** https://github.com/leechiehyu/ACMG-small_variant

Typical usage:

```
python 00_vep_batch_submitter.py <input_vcf_directory> <output_directory>
bash acmg_batch_submitter.sh
```

The annotation repository contains the batch submitter, per-chromosome preprocessing and VEP scripts, and the MANE Plus Clinical BED regions; the classification repository contains the rule modules (`acmg_rules.py`, `aggregation.py`, `data_loader.py`), the configuration (`config.json`, `hg38_BA1exc.tsv`), and the environment specification. Operational aids — a quick-start guide (commands, input formats, paths, outputs, and common error handling) and a tutorial recording — accompany the repositories, with issues tracked on GitHub and continued support via the GDK group.

7. Limitations & Future Work

- **Mitochondrial variants** are not yet fully supported; MT-specific handling is planned.
- **PM2 / BS2 thresholds** are currently global; disease-specific frequency thresholds are a planned refinement.
- **Decision-tree refinement** — several criteria warrant more granular sub-trees; some computational strengths (e.g., CADD-based evidence) could be extended toward higher strengths.
- **Version dependence** — results depend on reference genome, transcript set, and database versions (VEP v115, MANE v1.4, ClinVar 20251109); different transcripts can yield different HGVS and consequence.
- **Automation boundary** — clinical-evidence criteria (PS2, PS4, PM3, PP1, PP4, BS4, and related) require human review and are not automated.
- **PP5 / BP6 (reputable source)** are retained for coverage, though their use is debated because of potential double-counting; VUS is not a diagnosis, and classifications may change as databases update.
- **Scope** — the workflow is for germline small variants (SNV / indel / MNV); complex splicing, CNV, SV, and repeat-expansion loci require dedicated tools. Current use is research analysis and candidate filtering; clinical reporting still follows laboratory quality-management, validation, review, and sign-off procedures.

8. Availability & Versions

Table 10. Key resources and versions used in this workflow.

Resource	Version / release
Reference genome	GRCh38
Ensembl VEP	v115
MANE	v1.4
ClinVar	20251109
DeafnessVD (DVD)	v9.2
Population frequency	gnomAD, TWB, SG10K
Prediction / annotation	SpliceAI, dbscSNV, MaxEntScan, dbNSFP, PrimateAI, LoFtool, pLI, LOEUF, DosageSensitivity, NMD, satMutMPRA, CADD, DANN, RepeatMasker
Gene / MOI curation	CGD, ClinGen, PanelApp, GenCC, HGMD, MitoMap
Compute	NCHC / NARLabs

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Exact point thresholds for final classification are defined in `utils/config.json`; score/point adjustments are made in `utils/rescore.py`. Prepared from the NCHC workflow technical materials.